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PosturalNet, an Advanced Architecture to Detect Shoplifting at Megastores

Mohd Aquib Ansari
SCSET, Bennett University,
Greater Noida, India
mohd.aquib@gmail.com

Shahnawaz Ahmad
SCSET, Bennett University,
Greater Noida, India
shahnawaz.ahmad@bennett.edu.in

Arvind Mewada
SCSET, Bennett University,
Greater Noida, India
arvind.mewada@bennett.edu.in

Sushil Kumar Maurya
ITER, Siksha 'O' Anusandhan
University, Bhubaneswar, Odisha
sushilmaurya@soa.ac.in

***Abstract*— Human activity recognition (HAR) requires a full understanding of postural dynamics to create sophisticated surveillance systems that specifically detect shoplifting activities. PosturalNet represents a new architectural framework that uses human postural analytics to detect shoplifting incidents in large retail spaces. The proposed method uses a 2D body joint detection algorithm named OpenPose to extract postural information. Subsequently, these postures are pre-processed, and then features such as joint velocities, normalized body joints, magnitudes and angles are extracted. A combination of feature encoding techniques together with dimension reduction strategies maintains vital information while improving system computational efficiency. Testing PosturalNet on synthetic shoplifting data yielded an accurate detection rate of 96.87%. The method demonstrates robust performance in real-time applications which handle occluded environments when applied to practical store surveillance footage.**

Keywords— Human activity recognition, Video Surveillance, Machine learning, Deep learning, AI models.

I. INTRODUCTION

Large commercial stores throughout America, India, China, etc., experience massive annual financial losses from shoplifting. Advanced surveillance infrastructure with CCTV systems operates throughout retail facilities, yet shoplifting remains difficult to detect in real-time. Current video monitoring-based surveillance techniques demand personnel in control rooms to watch a non-stop stream of feeds throughout extended periods, ultimately leading to operational fatigue and monitoring deterioration. Current surveillance systems fail to detect critical events because of this passive failure mechanism, leading to ineffective theft prevention.

Human posture estimation systems and HAR technologies display promising outcomes for addressing obstacles stemming from variable body positions and environmental blocking. HAR systems have gained widespread adoption throughout security applications [2][3] and healthcare environments and human-computer interface applications because they extract features either through deep learning or engineered techniques from video sequences [1][4]. Detection methods based on these techniques struggle to differentiate activities in environments characterized by retail shop complexity and chaos.

Researchers enhance human posture estimation by combining robust algorithms such as OpenPose [6] along with methods from [5][7][8], which detect accurate body joints while generating geometric interpretations for human poses. Algorithmic performance based on posture analysis perfectly suits real-time video monitoring systems. Current approaches show constraints regarding real-time accuracy and occluded posture recognition in changing environments.

This paper presents an advanced 2D posture-based detection system called PosturalNet to detect shoplifting in large retail environments. OpenPose enables PosturalNet to extract skeletal features through body joint estimation. In the next phase, skeleton preprocessing occurs, where occlusion is handled by estimating joint positions over occluded joints. Further, such pre-processed skeletons are passed to the feature extraction module, where normalized key point data processing, joint velocity measurement, magnitude calculations and angle identification are estimated. The combination of Principal Component Analysis (PCA) alongside Gated Recurrent Units (GRU) enables proper classification through feature analysis. Our method beats existing systems for shoplifting detection by delivering an outstanding performance accuracy of 96.87%. The solution presents affordable scalability features that prove beneficial for retail store security improvement on a large scale.

The remainder of this paper is organized as follows: Section II addresses previous studies in this area while identifying the weaknesses in current solutions. The proposed method, including feature extraction and classification steps, is detailed in Section III. Section IV demonstrates how experimental results confirm the success of this method. Fifth, conclusions follow along with plans for future work in Section V.

II. RELATED WORK

The research field of Human Activity Recognition (HAR) has expanded across multiple decades to deliver essential progress toward secure systems and monitoring

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platforms and human-machine interactions. Real-time activity classification success has greatly improved through deep-learning architectures, including CNNs and RNNs, following the evolution of HAR systems from handcrafted features to advanced deep-learning architectures. The table compares prominent works from activity recognition while showcasing their respective methodologies.

Table 1. Summary of Related Works

Reference	Methodology	Key Contributions
[9]	Handcrafted features (HOG, GiST, color features) with Random Forest classifier	Recognized activities in multimodal egocentric videos
[10]	3D-CNN architecture	Encoded contextual information for activity recognition
[11]	CNN + LSTM (Deep Inception V3 for feature extraction and LSTM for classification)	Classified human acts in indoor surveillance
[12]	Xception + LSTM	Classified anomalies in synthesized video sequences
[13]	Recurrent Convolutional Networks	Learned spatiotemporal dynamics for large-scale visual understanding
[14]	2D pose estimation (OpenPose) with LSTM	Analyzed motion dynamics for action recognition
[15]	Multi-cue learning (lp-norm kernel combining motion, object, and scene semantics)	Recognized complex actions by fusing temporal and contextual information
[16]	Deep neural networks for 2D posture-based action recognition	Encoded body poses into representations and combined pose and global scores for activity labeling

The research shows how Handwear Action Recognition has progressed through HOG features, and GiST features [13] as well as deep learning frameworks comprising 3D-CNN [3] and RNN versions, including LSTM and GRU [4, 12, 15]. The proposed method showcases how integrating space and time elements creates successful activity recognition systems across multiple real-world applications, which include detecting anomalies [14], analyzing egocentric videos [13] and processing large data sets [15]. Modern 2D pose estimation methods represented by OpenPose [16] attain real-time joint detection, which assists in managing challenging occluded and dynamic environments. Farzan et al. [16] and Chenyang et al. [18] successfully used posture estimates to improve activity recognition precision. The current detection technologies struggle with two major problems: high computational requirements and limited application to shoplifting events in obscured settings. This work develops PosturalNet as a cost-effective system that uses human postures and complex posture analysis to identify shoplifting activities in unobstructed environments and areas where visibility is impaired.

PosturalNet surpasses existing HAR models by offering efficient, real-time, and occlusion-resilient activity recognition. Its parallel implementation of GRU and pose-based motion analysis lowers computational costs and suits real-world surveillance because it does not rely on deep learning architecture. PosturalNet surpasses traditional models by handling occlusions alongside dynamic environments through its ability to capture postural cues for detecting suspicious actions. The system excels above standard HAR frameworks through optimized shoplifting detection algorithms because it detects atypical behaviours with improved precision and recall values. The alignment of PosturalNet ensures strong operational ability in busy or hard-to-view retail scenarios, which is better than traditional CNN and recurrent-based techniques regarding both efficiency and accuracy rates.

III. PROPOSED METHOD

Through PosturalNet, researchers developed a method to analyze megastore security footage using data extracted from human posture and movement patterns. The methodology follows a structured pipeline, as depicted in Figure 1, which consists of four main stages: PosturalNet manages its analytical process across four staged operations, which start with Pose Generation and Skeleton Preprocessing and then move onto Feature Evaluation and Feature Reduction along with Classification.

A. Pose Generation

System processing converts video sequences received from CCTV cameras into human skeletal information. Input data landing at the OpenPose or compatible pose estimation algorithm produces recorded body joint coordinates through the pose generation module. The process establishes a precise portrayal of human motion patterns using skeletal information.

B. Skeleton Preprocessing

Once skeletal data is extracted, a preprocessing phase is applied to refine the input features by performing the following tasks:

(a) Discarding Irrelevant Frames: This preprocessing phase removes those skeletons that add no significant meaning to the analysis, such as empty or static frames.

(b) Selecting Relevant Joints: From OpenPose, we receive a human skeleton with a total of 18 body joints. The position of the head and legs matters little to shoplifting behaviours, and we eliminate five points from the head (head along with both ears and eyes) and two points from the legs (hips and knees) from the skeletal analysis. This feature size reduction, coupled with better computational power, becomes achievable through this change. At last, the final skeleton has nine body joints: one neck, two shoulders, two elbows, two wrists, and two hips.

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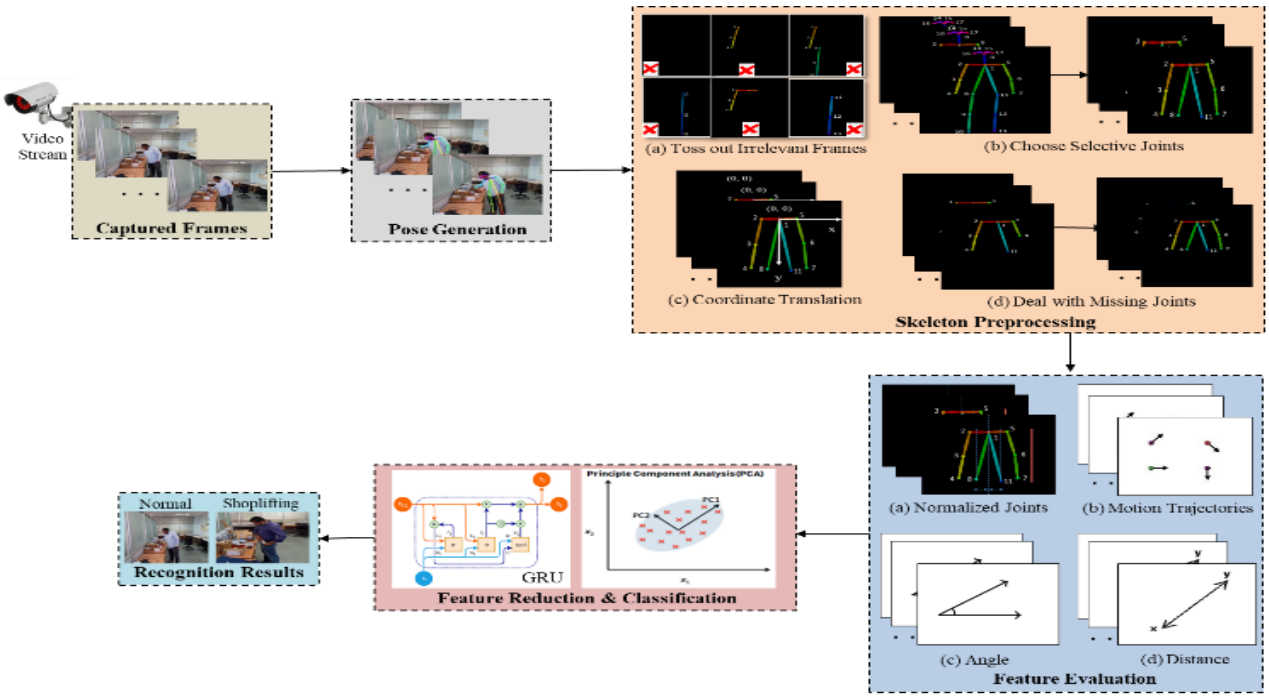


Fig. 1. Workflow of the Proposed Method

(c) Coordinate Translation: The normalization process divides coordinates against an origin point, resulting in scale-invariant outputs. Lets $\boldsymbol{S} = \{\boldsymbol{x_0}, \boldsymbol{y_0}, \boldsymbol{x_1}, \boldsymbol{y_1}, \boldsymbol{x_2}, \boldsymbol{y_2} \ \ldots, \boldsymbol{x_i}, \boldsymbol{y_i}\}$ where S represents the skeleton and $\{\boldsymbol{x_0}, \boldsymbol{y_0}, \boldsymbol{x_1}, \boldsymbol{y_1}, \ \ldots, \boldsymbol{x_i}, \boldsymbol{y_i}\}$ are called body joints.

$$px0, py0 = get_{joint(S,NECK)} \quad (1)$$

$$S[0 :: 2] = S[0 :: 2] - px0 \quad (2)$$

$$S[1 :: 2] = S[1 :: 2] - py0 \quad (3)$$

Here, px0 and px1 are the coordinates for the neck joint. Each body joint receives its position through scaling based on the neck joint values subtracted from each coordinated value.

(d) Handling Missing Joints: The system reconstructs missing or obstructed joints by utilizing information from surrounding limb positions while blending this data with the previous skeleton frame points.

$$x_i = x_0 + x_i_prev \quad (4)$$

$$y_i = y_0 + y_i_prev \quad (5)$$

Here $(\boldsymbol{x_0}, \boldsymbol{y_0})$ is the spatial location of the neck joint in the current frame and $(\boldsymbol{x_i_prev}, \boldsymbol{y_i_prev})$ is the set of the spatial locations of the missing joints in the previous frame.

C. Feature Evaluation

Following preprocessing operations, the processed skeletal information transforms into distinguishable features before classification. Let $\boldsymbol{X} = \{\boldsymbol{S_0}, \boldsymbol{S_1}, \boldsymbol{S_2}, \boldsymbol{S_3}, \ \ldots, \boldsymbol{S_N}\}$, where X is the set of skeletons for N frames. Each point in the skeleton is represented by a pair of spatial coordinates $(\boldsymbol{x_i}, \boldsymbol{y_i})$. The following feature representations are computed:

(a) Normalized Joints: The coordinates tackle spatial normalization to preserve participant consistency.

$$Feat_1 = \frac{S_j}{Mean(Height\ H\ of\ each\ skeleton)} \quad (6)$$

$$Where\ 0 \leq j \leq N$$

(b) Motion Trajectories: Temporal movement patterns of key joints are calculated over consecutive frames as follows.

$$\boldsymbol{Feat_2 = X[S_N] - X[S_{N-1}]} \quad (7)$$

(c) Joint Angles: Body part angle deviation measures movement dynamics between human components.

$$\boldsymbol{Feat_3 = \tan^{-1} \frac{(x_j - x_k)}{(y_j - y_k)}} \quad (8)$$

(d) Distance Features: The algorithm measures pose deviations by calculating Euclidean distances between crucial joints.

$$\boldsymbol{Feat_4 = \sqrt[2]{(x_k - x_j)^2 + (y_k - y_j)^2}} \quad (9)$$

Table 2. Summarized features

Features	Measurement	No. of Features
Normalized key points ($Feat_1$)	9 key points × 2 coordinate/key points × (N) images	2610

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Joints Velocity ($Feat_2$)	9 key points × 2 coordinate/key points × (N - 1) images	2592
Joints Magnitude ($Feat_3$)	9 key points × 2 coordinate/key points × (N - 1) images	2592
Joints Angle ($Feat_3$)	10 Angles × N images	1450
Final Features (F) = [$Feat_1$, $Feat_2$, $Feat_3$, $Feat_4$]		9244

In Table 2, using human skeletal data, the proposed method obtains 9,244 features which encode shoplifting activities. The features utilize normalized key points between all frames while delivering 2,610 features which track the location of 9 essential joint points. The method analyzes motion patterns through the calculation of joint velocity and joint magnitude, which generate 2,592 features each between consecutive frames. The analysis utilizes joint angles from individual frames to understand postural variations using 1,450 measurement points from 9 key joint measurements. The robust representation for classification results from combining extracted parameters into the final feature set [$Feat_1$, $Feat_2$, $Feat_3$, $Feat_4$].

D. Feature Reduction & Classification

PCA reduces the high-dimensional feature set from raw skeletal input while maintaining discriminative features since skeletal data contains many dimensions. The reduced feature set is then passed to a Gated Recurrent Unit (GRU)-based deep learning model, which captures temporal dependencies in motion patterns and classifies sequences into two categories: Normal Activity and Shoplifting Behavior.

E. Recognition & Alert System

Real-time processing outputs from classification lead to recognition results, while an integrated alert system allows security personnel to receive shoplifting detection warnings.

IV. EXPERIMENTAL OUTCOMES

The hardware uses an Intel i5 processor, a fast 500GB SSD, and an NVIDIA GTX 1650 GPU running under 8GB RAM to deliver smooth operation and efficient computations. Real-time processing capabilities combined with intelligent surveillance insights became possible through the system's 8GB RAM capacity that handled the shoplifting behavior analysis without interruption. The hardware setup found its optimal speed-efficiency ratio, allowing deep learning models to perform precise anomaly detection during training sessions. This combination of advanced processing power with optimized storage capabilities enabled analysis of each video frame at astonishing speed to redefine smart security possibilities.

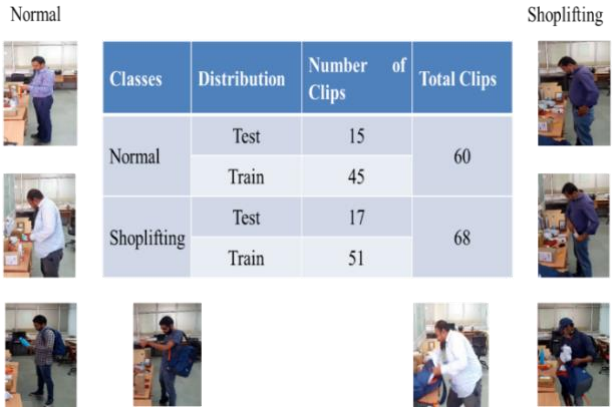


Fig. 2. Shoplifting dataset distribution

Figure 2 showcases a dataset designed for shoplifting detection, comprising video clips categorized into two main classes: Normal and Shoplifting. Our dataset includes separated training and testing sections, enabling machine learning model training and subsequent evaluation procedures. The Normal class contains sixty clips distributed between forty-five training and fifteen testing sequences. The Shoplifting class spans 68 clips, containing 51 training and 17 testing elements. The distribution arrangement displays its structure through this central table. Surrounding the table, several sample images illustrate various activities captured in the dataset. Some images depict individuals engaged in normal shopping behaviour, such as browsing or interacting with items. In contrast, others show suspicious actions associated with shoplifting, such as concealing items or quickly exiting the scene.

Table 3. Experimental Results

Confusion Metrix		Measures	Values
14 (TP)	1 (FN)	Positive Predictive Value (PPV)	1.000
		True Positive Rate (TPR)	0.933
0 (FP)	17 (TN)	Accuracy	96.87%

Evaluation results confirm that the proposed shoplifting detection model reaches a high detection accuracy of 96.87% to differentiate normal and suspicious behaviours. The Positive Predictive Value (PPV) indicated that all shoplifting cases identified by the model were accurate because they produced no wrong positive predictions. Exceptions to actual shoplifting events are rare because the model achieves a True Positive Rate of 0.933 yet suffers just one False Negative instance (See Table 3). Observations categorized as true negatives received proper classification from the model, demonstrating its adaptability for actual security surveillance operations.

A comparison between different detection classifiers appears in Figure 3. It compares the outcomes of different classifiers in the presence of PCA and in the absence of PCA. All classifiers demonstrated pretty good accuracy when Principal Component Analysis (PCA) was applied to the analysis. The proposed approach reaches a maximum accuracy of 96.87% when implemented with a GRU (Gated

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Recurrent Unit) classifier. The study demonstrates that SVM and RF classifiers alongside Principal Component Analysis (PCA) yield significant performance enhancements, demonstrating PCA's dual ability to reduce features while improving discrimination capabilities.

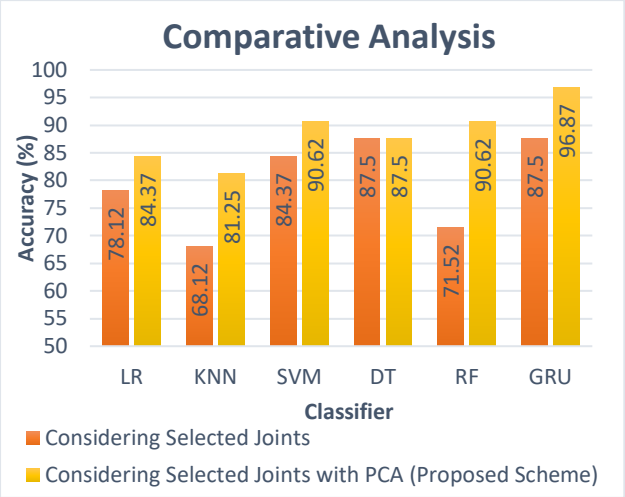


Fig. 3. Comparative analysis to show the effectiveness of proposed method comprising without PCA and with PCA

Table 4. Comparative Analysis

Methods	Jayaswal and Dixit [12]	Ansari and Singh [11]	Ansari and Singh [1]	Proposed Method
Accuracy	87.23%	89.36%	91.48%	96.87%

The proposed method delivers higher accuracy at 96.87% than the two existing methods, which secured 87.23% and 89.36%, and 91.48% success rates, respectively, in Table 4. The proposed framework performs better through optimized pose-based feature extraction to detect shoplifting incidents.

V. CONCLUSION

This work showcases PosturalNet, which represents an enhanced pose-based deep learning system for detecting shoplifting events in megastores. The proposed detection framework extracts postures from skeleton-based human position detection methods to create motion and spatial features that proceed through dimensional reduction and then move on to classification assessment. Experimental findings indicate that the model reaches high accuracy at 96.87% while surpassing all previous methods. The system achieves outstanding performance in detecting shoplifting activities with a True Positive Rate of 0.933 and a Positive Predictive Value of 1.000. The proposed technique stands superior to previous methods in comparison testing. The findings show that this pose-based action recognition system offers an efficient and economical solution that solves video surveillance obstacles related to human posture adaptations, background interference, and object blocking. Upcoming research will concentrate on real-time system optimization while expanding current data collection to enhance performance and implementing multi-camera tracking to increase reliability for dense store environments. A system for shoplifting detection will achieve significant improvements through the combination of depth and thermal and LiDAR sensor data with RGB video enabled by advanced deep learning models such as ViTs and ST-GNNs to boost detection reliability across different retail store layouts.

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